

Problem H

Fridgeberg's Challenge

Time limit: 1 second

Problem Description

Fridgeberg is a well-known veteran in the Taiwanese ICPC community. He specializes in counting problems. Since he solves counting problems faster than everyone else, people call him a speed-counter. Today, he has prepared a challenge for you along with the following message.

“Yo, I’m Fridgeberg, and I’m an 18-year-old ICPC speed-counter from Yunlin, Taiwan. Today, I’m challenging you to a counting problem.”

The challenge is as follows.

A rooted labelled tree with n vertices is a tree with vertices labelled $1, 2, \dots, n$ and one vertex chosen as the root.

Recall that for any two vertices in a tree, there is a unique simple path connecting them. For each vertex v in a rooted labelled tree, define its subtree as the set of all vertices u such that v is on the unique path from the root to u . In particular, a vertex is always in its own subtree. The subtrees of a tree are the collection of all vertices’ subtrees.

A rooted labelled tree is k -free if none of its subtrees has exactly k vertices.

For each k from 2 to $n - 1$, compute the number of different rooted labelled trees with n vertices that are k -free. Since the numbers can be very large, please compute them modulo 998244353.

Two rooted labelled trees are different if at least one of the following holds:

- The labels of the roots are different.
- There exists an edge that is present in one tree but not in the other.

Input Format

The only line contains a single integer n .

Output Format

Output $n - 2$ lines. On the i -th line, output the answer when $k = i + 1$.

Technical Specification

- $3 \leq n \leq 2 \times 10^5$

Sample Input 1

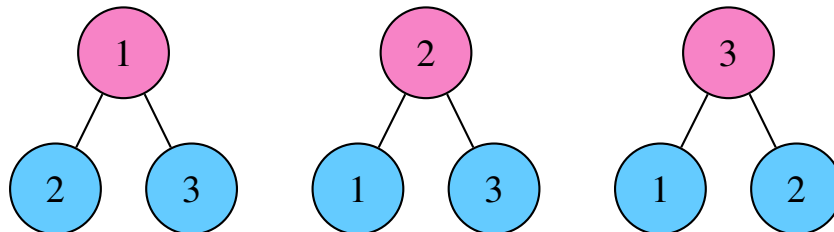
3

Sample Output 1

3

Explanation for Sample 1: There are exactly 3 rooted labelled trees that have no subtree with exactly 2 vertices.

The following picture illustrates those 3 trees. (The root vertices are colored pink, and the other vertices are colored light blue.)



Sample Input 2

4

Sample Output 2

16
28

Sample Input 3

11

Sample Output 3

23790225
 701354623
 477759311
 29734262
 735403659
 26527433
 622702793
 495236333
 962003659